DAVID MDLULI

Final-Year Information Technology Student
Aspiring Software Engineer | Junior Software Developer
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Final-year BSc IT student at North-West University, pursuing a career in software engineering. Hands-on experience developing applications using C#, .NET, Java, Python and SQL, including database-driven systems, mobile applications and automation tools. Experienced in implementing CRUD functionality, database integration and object-oriented programming, with additional exposure to layered architecture and unit testing. Currently serving as a Python and C++ Student Assistant, supporting over 100 students with programming concepts and debugging. Seeking graduate software engineering and junior software developer opportunities.

TECHNICAL SKILLS

Programming Languages: C#, Java, Python, C++, SQL
Frameworks & Technologies: .NET, .NET MAUI, ASP.NET Web Forms, JDBC, Maven, JUnit 5
Databases: SQL Server, SQLite, LocalDB
Data & AI: Pandas, NumPy, Matplotlib, scikit-learn, Streamlit
Tools: Git, GitHub, Visual Studio, VS Code
Software Engineering: OOP, CRUD Operations, Database Integration, Layered Architecture, Unit Testing, Debugging

PROFESSIONAL EXPERIENCE

STUDENT ASSISTANT (DEMONSTRATOR) - (PYTHON & C++)

NORTH-WEST UNIVERSITY | February 2025 - Present

Key Responsibilities

- Mentor more than 100 first-year students during Python and C++ programming practical sessions
- Guide students through debugging, programming logic and object-oriented programming concepts
- Support lecturers with practical assessments, marking and invigilation
- Appointed across four academic teaching periods between 2025 and 2026

PYTHON DEVELOPMENT INTERN

HEX SOFTWARES | April 2025 – May 2025

- Developed a Python voice-controlled virtual assistant to process spoken commands and automate desktop tasks
- Built a web scraper using Requests and BeautifulSoup to extract book information across 50 catalogue pages

EDUCATION

Bachelor of Science in Information Technology | North-West University
Expected Academic Completion: November 2026

RELEVANT COURSEWORK

- Object-Oriented Programming
- Systems Analysis & Design
- Data Structures & Algorithms
- Databases
- Computer Networks
- Artificial Intelligence
- Decision Support Systems

SOFTWARE PROJECTS

PERSONAL FINANCE MANAGER (GITHUB)

Technologies: Java, SQLite, JDBC, Maven, JUnit 5

- Developed a Java console application for managing income, expenses, and personal financial transactions
- Structured the application using layered architecture and implemented CRUD operations with SQLite persistence through JDBC
- Implemented transaction validation, automated financial calculations and JUnit 5 tests to verify application behaviour

STUDENT TASK MANAGER - MOBILE APPLICATION (GITHUB)

Technologies: C#, .NET MAUI, SQLite, CommunityToolkit

- Independently developed a cross-platform mobile application for organising academic tasks
- Implemented user registration, login and CRUD operations for task management
- Integrated SQLite for persistent local data storage and built a responsive user interface using .NET MAUI

HOPE THROUGH HANDS - CHARITY MANAGEMENT SYSTEM (GITHUB)

Technologies: ASP.NET Web Forms, C#, SQL Server

- Collaborated on backend development for a charity management system supporting donors, beneficiaries and donations
- Implemented CRUD functionality, input validation and application business logic using C#
- Integrated application functionality with SQL Server and used GitHub for collaborative development

STUDENT PERFORMANCE ANALYTICS (GITHUB)

Technologies: Python, Pandas, NumPy, Matplotlib, scikit-learn, Streamlit

- Explored relationships between study time, absences, previous failures and academic performance
- Applied logistic regression to develop a student performance prediction model and present analytical insights

GIBBS - PYTHON VIRTUAL ASSISTANT (GITHUB)

Technologies: Python, OpenAI, SpeechRecognition, pyttsx3, pywhatkit

- Built a rule-based voice assistant with speech recognition and text-to-speech for spoken interactions
- Integrated OpenAI for AI-powered responses and interaction
- Automated tasks such as YouTube searches, weather queries and desktop actions

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CERTIFICATIONS

MTN SKILLS ACADEMY

- AI Basics
- Generative AI

FNB APP ACADEMY

- Certificate in Full Stack Development
